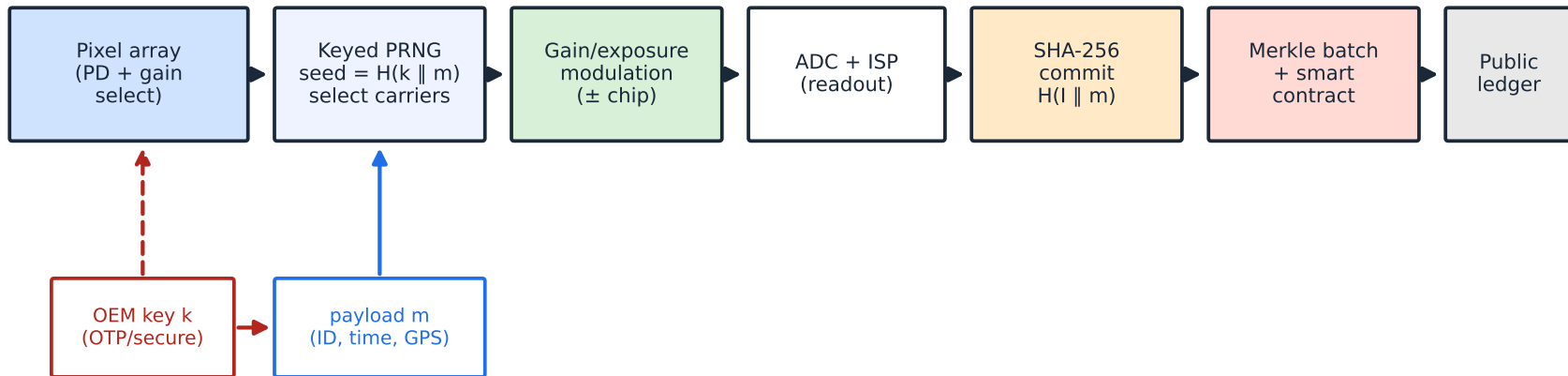


PixelRoot provenance-write datapath (capture-time)



Soft binding (in-pixel, survives transcoding) + Hard binding (on-chain hash) — two independent barriers.